



Institutions, Trade & Economic Growth

Days 2-3: Coordination problems and institutions

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King's College London

Summer School

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Coordination Problems & Institutions

Questions we will ask:

- When do economic interactions produce desirable aggregate outcomes? (like inclusive economic growth or technological progress)
- And when do they lead to societal outcomes that no one would have desired? (like underdevelopment or excessive pollution)
- How can we represent economic interactions in a stylized way to study them & predict their outcomes?
- What are institutions and how can they solve (or worsen) coordination problems?



The plan for days 2-3

1. Coordination Problems
2. Games
3. Pareto efficiency
4. Types of Games
5. Institutions and Coordination Problems

The main ideas we will explore

- **Coordination problems**: decentralised action can produce outcomes that are bad for society
 - inefficient production, underdevelopment, excessive environmental damage, ...
- **“Classical institutional challenge”**: setting rules of the game that leave individuals free to choose their own actions but deliver outcomes that are good for society.
- The ‘rules of the game’ are also called **institutions**:
 - *The laws, informal rules, and mutual expectations which regulate social interactions among people and between people and the biosphere.*
- Institutions determine which actions are available to economic agents, and the incentives they face.
- Institutions determine whether coordination problems are solved.

Our intellectual guides



David Hume
1711-1776

*"Two neighbors may agree to drain a meadow, which they possess in common...
...But 'tis very difficult and indeed impossible, that a thousand persons shou'd agree in any such action"*



Adam Smith
1723-1790

*"He intends only his own gain, and he is in this, as in many other cases, led by an invisible hand to promote an end which was no part of his intention. ...
By pursuing his own interest he frequently promotes that of the society more effectually than when he really intends to promote it."*

Palanpur farmers



They sow their crops several weeks after the date at which they would maximize crop yields.

Why?

"...Palanpur farmers sow their winter crops several weeks after the date at which they could maximize crop yields.

The farmers do not doubt that earlier planting would give them larger harvests, but no one is willing to be the first to plant their seeds, as the seeds on any lone plot would be quickly eaten by birds.

I asked if a large group of farmers, perhaps relatives, had ever agreed to sow their seeds earlier, all planting their seeds on the same day to minimize the losses. "If we knew how to do that, [the farmer] said, we would not be poor".

(from Bowles & Halliday, 2022)

Rousseau's stag hunt

- *The stag hunt will succeed only if everyone stays at their place.*
- *But if a participant sees a hare, she will leave her place to catch it, making the stag hunt fail.*



Rousseau's stag hunt

"Were it a matter of catching a deer, everyone was quite aware that he must faithfully keep to his post in order to achieve this purpose; but if a hare happened to pass within reach of one of them, no doubt he would have pursued it without giving it a second thought, and that, having obtained his prey he cared very little about causing his companions to miss theirs."

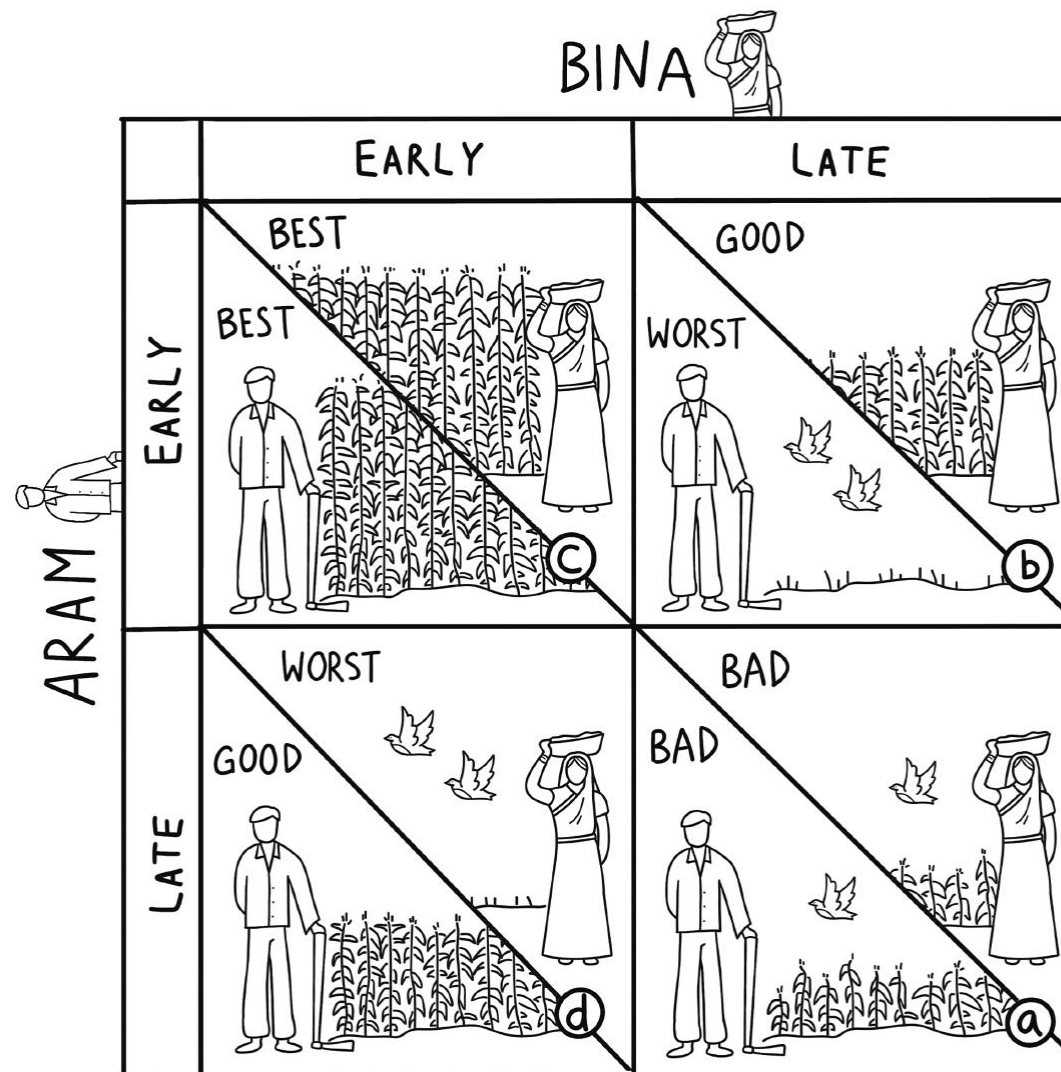
Jean-Jacques Rousseau (1755)

COORDINATION FAILURES: people trying to do their best individually can collectively produce inefficient, undesirable outcomes, because they don't coordinate their actions.

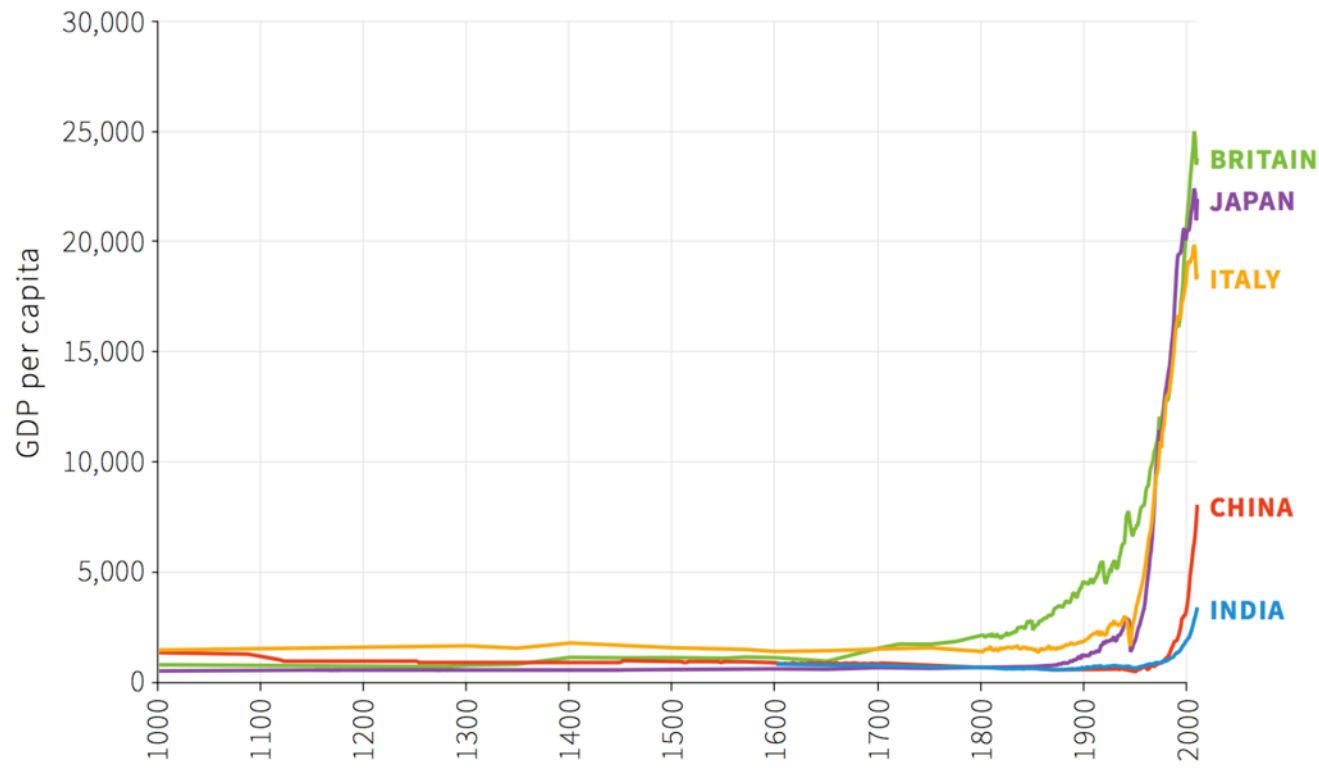
Can you think of some examples (big or small)?

CLASSICAL INSTITUTIONAL CHALLENGE:

- How to get the “rules of the game” right?
- How to design *institutions* such that people are free to choose their own actions while at the same time avoiding undesirable collective outcomes? (*David Hume, Jean Jacques Rousseau*)



Are coordination failures important in our society?



Are coordination failures important in our society?

- **On the one hand**, under capitalism, individuals acting mostly independently, seeking their personal objectives, have produced an unprecedented long-term increase in average living standards (the *invisible hand*).
- **On the other hand**, uncoordinated activities have also produced outcomes that are bad for everyone (*coordination problems*):
 - Overuse of natural resources
 - Underdevelopment in many countries
 - Excessive pollution, climate change
 - Unemployment (underuse of human resources)
 - Financial Crises
 - Underuse of vaccines - overuse of antibiotics
 - Teamwork that fail because everyone tries to just rely on the effort of others.

Why do coordination failures arise?

Externalities → Coordination Failures

- **Coordination failures** have a common root: **externalities**
- Externalities (or *external effects*) occur when an individual's actions generate costs or benefits for other people who are not part of the decision.
 - Individuals affect the well-being of others but do not take these effects into account when making their choices.
- **Internalizing externalities** means ensuring that people take the effects of their actions on others into account in making choices.
- When externalities are fully internalized, coordination problems disappear
- Easier said than done!

Coordination Failures and externalities



- **Negative externalities** impose costs on others
 - Stag hunt: Leaving one's post to catch a hare reduces everyone else's chance of catching the stag.
 - Planting in Palanpur: Planting late increases the risk that neighbors' seeds are eaten by birds.
 - Climate change: Emitting greenhouse gases contributes to global warming, which affects people all around the world.
- **Positive externalities** provide benefits to others
 - Vaccination: Getting vaccinated reduces disease transmission, benefiting others who might otherwise become infected.
 - Teamwork: Working hard benefits the whole team, including those who contributed less (*free riding* problem).
- If not internalized, both create coordination failures.

Institutions & Coordination Failures

- Institutions determine our ability to solve coordination problems.
- Institutions: the laws, informal rules, and mutual expectations which regulate social interactions among people and between people and the biosphere.
- Institutions are the *rules of the game*:
 - They determine which actions are available to individuals, and the associated incentives.
- Game theory can be used to understand how institutions (the rules of the game) affect the outcome of strategic interactions.

Games

Game Theory models social interactions as strategic games.

In a game:

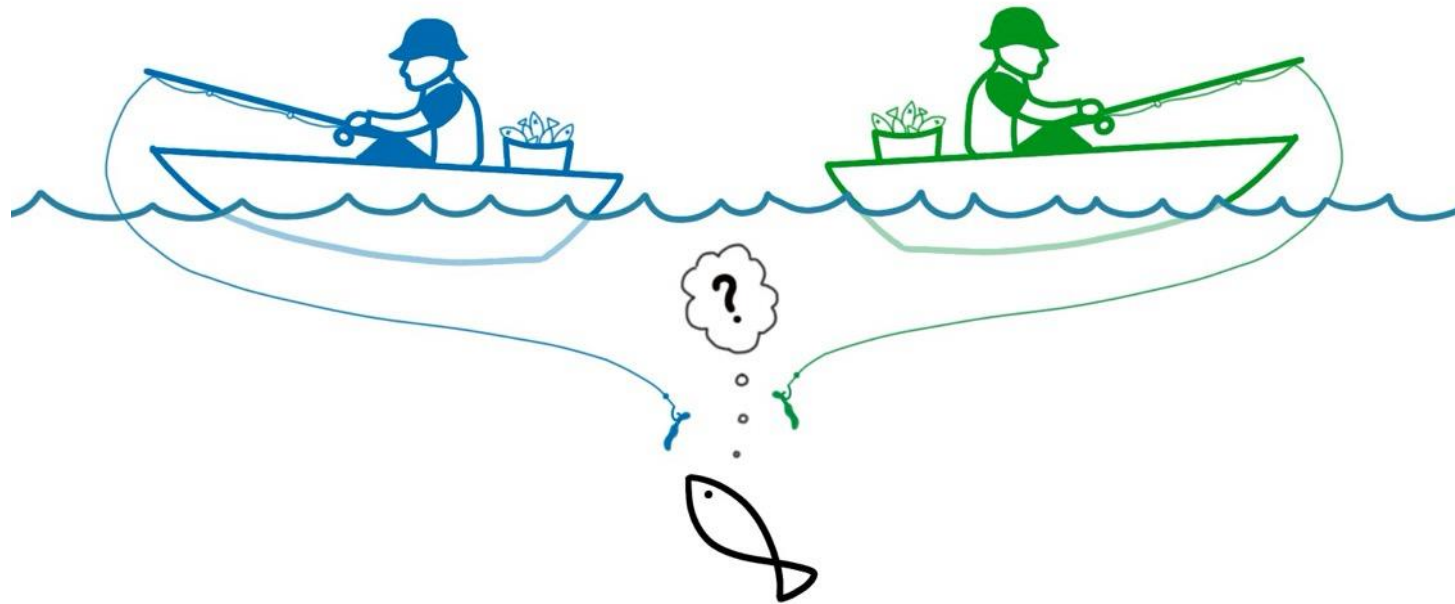
- each player has alternative actions available.
- each player's payoff depends on both her own actions and other players' actions.
- *Interdependence*: what you get depends not only on what *you* do, but also on what *others* do [*social* interaction].
- *Institutions* (“the rules of the game”) are the underlying norms that determine which actions are available to whom, and who gets what in each possible outcome.

Games

The basic ingredients of a **game**:

- **Players**
- **Strategy Set** (the actions available to players)
- **Payoffs** (what players get in each possible combination of actions, or *strategy profile*)
- **Order of play** (simultaneous or sequential)
- **Information structure** (what do players know)

The Fishermen's Dilemma



The fishermen's dilemma

Alfredo & Bob share a lake for fishing.

As one catches more fish, the other catches fewer (an externality!)

Each must decide how long to fish for: **10** or **12** hours

Fishing longer costs time & effort, but yields more fish.

If both fish 12 hours, they each catch fewer fishes per hour, while spending a lot of time and effort → Each gets a payoff of 2

If they both fish 10 hours, they spend less hours but catch more fishes per hour → Each gets a payoff of 3

If Alfredo fishes 10 hours and Bob 12 hours, A gets few fishes while B gets a lot → Alfredo gets 1, Bob gets 4.

If Alfredo fishes 12 hours and Bob 10 hours → Alfredo 4, Bob 1.



A strategic (or normal) form game: The fishermen dilemma

		Bob	
		Fish 10 hours	Fish 12 hours
Alfredo	Fish 10 hours	3, 3	1, 4
	Fish 12 hours	4, 1	2, 2

- If they both fish 10 hours, what is Bob's payoff?
- If Alfredo fishes 10 and Bob fishes 12, what is Alfredo's payoff?
- If Alfredo fishes 12 and Bob fishes 10, what is Bob's payoff?

		Bob	
		Fish 10 hours	Fish 12 hours
Alfredo	Fish 10 hours	3, 3	1, 4
	Fish 12 hours	4, 1	2, 2

		Bob	
		Fish 10 hours	Fish 12 hours
Alfredo	Fish 10 hours	3, 3	1, 4
	Fish 12 hours	4, 1	2, 2

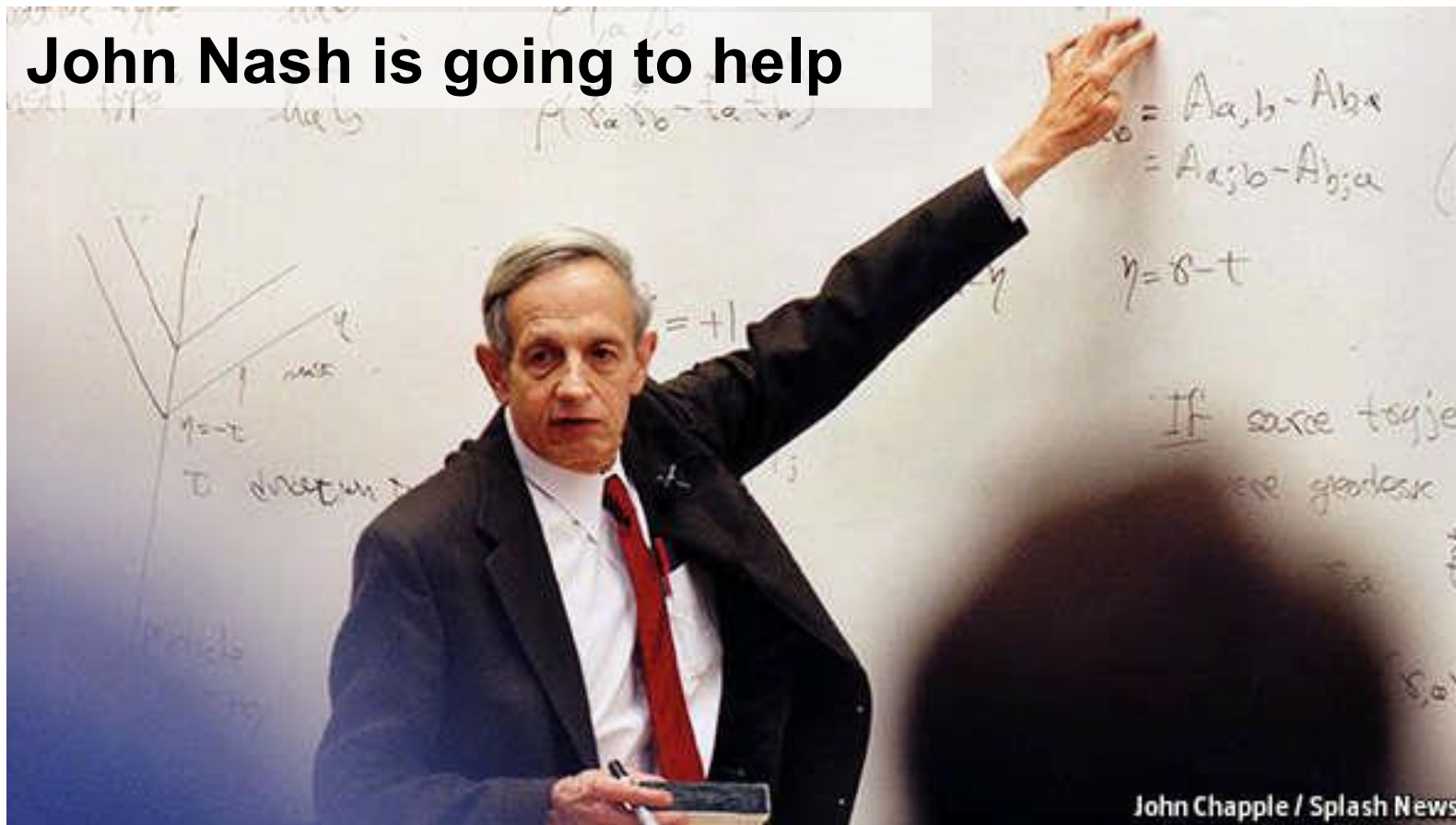
If both fish 10h, they both get a payoff of 3

If Al fishes 10h and Bob fishes 12h, Al gets 1 and Bob gets 4

If Al fishes 12h and Bob fishes 10h, Al gets 4 and Bob gets 1

If both fish 12h, they both get a payoff of 2

How do we **predict the outcome** of a game?



➤ Best responses:

the *best response of player A to strategy X of player B* is the strategy that gives A the highest payoff when B chooses strategy X.



➤ Nash Equilibrium

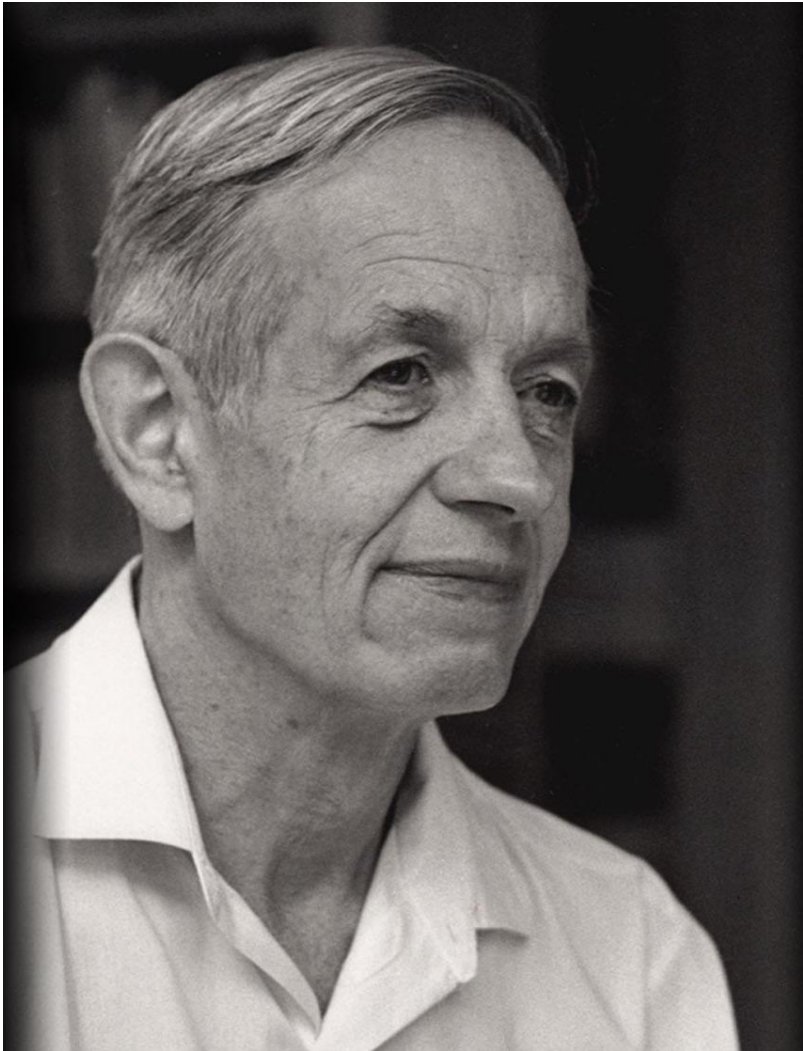
Equilibrium means that no one could do better by individually changing strategy (*stationary* state).

In equilibrium, everyone is playing their *best response* given the other players' actions.

➤ The solution of a game must be a Nash Equilibrium.

If this was not the case, that outcome would never be how the game ends!

Solving a game for its Nash equilibrium



3 steps to find the *Nash Equilibrium*:

1. Find the **best responses of A**, given each possible strategy of B.
2. Find the **best responses of B**, given each possible strategy of A.
3. The strategy profiles in which **both players are 'best responding'** to each other's strategies are the Nash Equilibria
 - (may be one or more).

1 – **Alfredo's best responses** to each possible action of Bob:

		Bob	
		Fish 10 hours	Fish 12 hours
Alfredo	Fish 10 hours	3, 3	1, 4
	Fish 12 hours	<u>4</u> , 1	<u>2</u> , 2

If B fishes 10h, what is A's best response?

If B fishes 12h, what is A's best response?

If Bob fishes 10h, Al's best response is to fish 12h (because $4 > 3$)

If Bob fishes 12h, Al's best response is to fish 12h (because $2 > 1$)

2 – Bob's *best responses* to each possible strategy of Alfredo.

If A fishes 10h, what is B's best response?

If A fishes 12h, what is B's best response?

Alfredo

Bob

	Fish 10 hours	Fish 12 hours
Fish 10 hours	3, 3	1, <u>4</u>
Fish 12 hours	<u>4</u> , 1	<u>2</u> , <u>2</u>

If A fishes 10h, Bob's best response is to fish 12h (because $4 > 3$)

If A fishes 12h, Bob's best response is to fish 12h (because $2 > 1$)

3 – What is the Nash Equilibrium?

		Bob	
		Fish 10 hours	Fish 12 hours
Alfredo	Fish 10 hours	3, 3	1, <u>4</u>
	Fish 12 hours	<u>4</u> , 1	<u>2</u> , <u>2</u>

(12,12) is a N.E.:
Both are playing a best response given the strategy of the other.

The Nash equilibrium

- A Nash equilibrium is a situation in which no player has a reason to change her strategy, as long as the other players do not change theirs.
- No one can increase their payoff by individually changing strategy.
- Mutual best response: The strategy of each player is a best response given the strategies of the other players.
- *Everyone is doing the best they can, given the actions of the others.*

The Nash equilibrium

Two important caveats:

- The definition of N.E. leaves unexplained *how the equilibrium will be reached*. We just know that once there, no one has an incentive to move away.
- A N.E. is *not* guaranteed to be a ‘good’ solution.

Dominant strategy equilibrium

- Sometimes, there is a single strategy that always yields the highest payoff to a player, no matter what the other player chooses.
- **Strictly dominant strategy**: it is the player's best response to all possible choices of the other player(s).
- If each player has a strictly dominant strategy, we have a **dominant strategy equilibrium**.
- In a **dominant strategy equilibrium**, each player has a unique strategy that is optimal, no matter what the others do (little interaction).
- A dominant strategy equilibrium is always a N.E.
- But a N.E. does not need to be a dominant strategy equilibrium.

Is the Nash Equilibrium of the **fishermen dilemma** a dominant strategy equilibrium?

Yes! For both players, fishing 12 hours is the best response to all possible actions of the other player.

		Bob	
		Fish 10 hours	Fish 12 hours
Alfredo	Fish 10 hours	3, 3	1, <u>4</u>
	Fish 12 hours	<u>4</u> , 1	<u>2</u> , <u>2</u>

Your turn! Find the N.E. of this game

A and B are two thieves that have been caught by the police. They are interrogated separately.

Player A

Snitch

Keep quiet

Player B

Snitch

Keep quiet

	Snitch	Keep quiet
Snitch	1, 1	3, 0
Keep quiet	0, 3	2, 2

- If they both snitch on each other, they both get 10yrs of jail (payoff=1).
- If one snitches and the other stays silent, the snitch gets free (payoff=3), the other goes to jail for 20yrs (payoff=0).
- If both keep quiet, they get away with just six months of prison each (payoff=2).

(Snitch, Snitch) is the Nash Equilibrium

		Player B	
		Snitch	Keep quiet
Player A	Snitch	<u>1</u> , <u>1</u>	<u>3</u> , 0
	Keep quiet	0, <u>3</u>	2, 2

Prisoners dilemmas & coordination problems

- The game you solved is a classic game called the ‘prisoners dilemma’.
- Fishermen dilemma & prisoners dilemma have something in common:

rational self-interest by each player produces an outcome that all players would have preferred to avoid → **Coordination failure**

- Is there any externality in these two games?

Does the N.E. provide a good description of how these kind of *strategic interactions* will unfold in reality?

- Often, it does.
 - especially if players don't have ways to coordinate their actions and don't care about each other.
- But not always!

Possible problems with the N.E.

It overlooks opportunities to coordinate

Imagine the two fishermen could sign a contract agreeing to fish 10h each, with a sizable penalty if one breaks the contract. Two rational fishers would be eager to sign that contract.

It assumes self-interest

If Alfredo and Bob cared about each other, fishing 10h each would be a sustainable equilibrium: none would increase their fishing hours, because none would want to damage the other.

There may be multiple equilibria

In some games, you can have more than one NE. Which one should we expect to be realized?

It ignores dynamics

Does not tell us how the equilibrium would be reached. But sometimes that's the main thing we would like to know.

What makes cooperation difficult in a coordination problem?

- Information: how do you check if the other is respecting the agreement? One could agree to cooperate but actually defect and get the biggest payoff!
- Enforcement: even if you can check on the other, how do you force him to comply?
- Distribution of benefits: Cooperating yields some extra-benefits. How to distribute them? *Conflict over distribution can derail cooperation*

→ *in situations like the Fishermen Dilemma or the Prisoners' dilemma, the Nash Equilibrium can often prevail and persist, even though it's bad for everyone.*

Coordination failures and economic growth

Coordination failures can hinder economic development

Two significant examples:

- Failure to unlock the gains from trade
- Failure to invest



Application: Unlocking the gains from trade

One obstacle from exploiting the gains from international trade is the resistance of sectors or groups that would lose from international competition.

These could be compensated using part of the gains from trade, but coordination problems can make the cooperative solution hard to achieve.

A simple game to capture this mechanism:

- Country with 2 industries must decide whether to open to international trade.
- Will open only if both industries vote in favor (veto power).
- Trade brings net benefits to the country.
- But distributed unequally: an internationally competitive 'Trade Winner' (TW) industry gains, while a less competitive 'Trade Loser' (TL) industry loses.
- After gains from trade have been realized, it is possible to redistribute them using taxes and subsidies, *if both industries vote in favor*.

Application: Unlocking the gains from trade

We could try write this as a simultaneous game...

TW

TL

Allow Trade

Veto trade

Allow
redistribution

Veto
redistribution

2, 2

-1, 5

0, 0

0, 0

But this misses an important dimension related to *timing*:

Redistribution can only happen *after* trade has been opened and the gains from trade realized (otherwise there is nothing to redistribute!)

Application: Unlocking the gains from trade

This is a *sequential* game:

1. Trade Loser decides whether to allow openness to trade or veto it.
2. If country opens & gains from trade are realized, Trade Winner decides whether to allow redistribution or veto it.

3 possible outcomes:

- No trade: $(0,0)$
- Trade and no redistribution: $(-1,5)$
- Trade and redistribution: $(2,2)$

Sequential games

- Players make decisions in a specific order.
- Unlike simultaneous games, later movers can observe the actions of earlier movers before making their choices.
- **Order of play** and timing can be critical to the outcome.
- Represented through a **game tree** (or “extensive form”)
 - Read from left to right (or sometimes from top to bottom)
 - **Nodes** represent points of decisions or final outcomes (the *terminal* nodes)
 - **Branches** represent the choices a player can make at each decision node.
 - **Payoffs** are listed at the terminal nodes that represent possible outcomes

🌐 Unlocking the Gains from Trade

A Sequential Game Illustrating Trade Coordination Problems



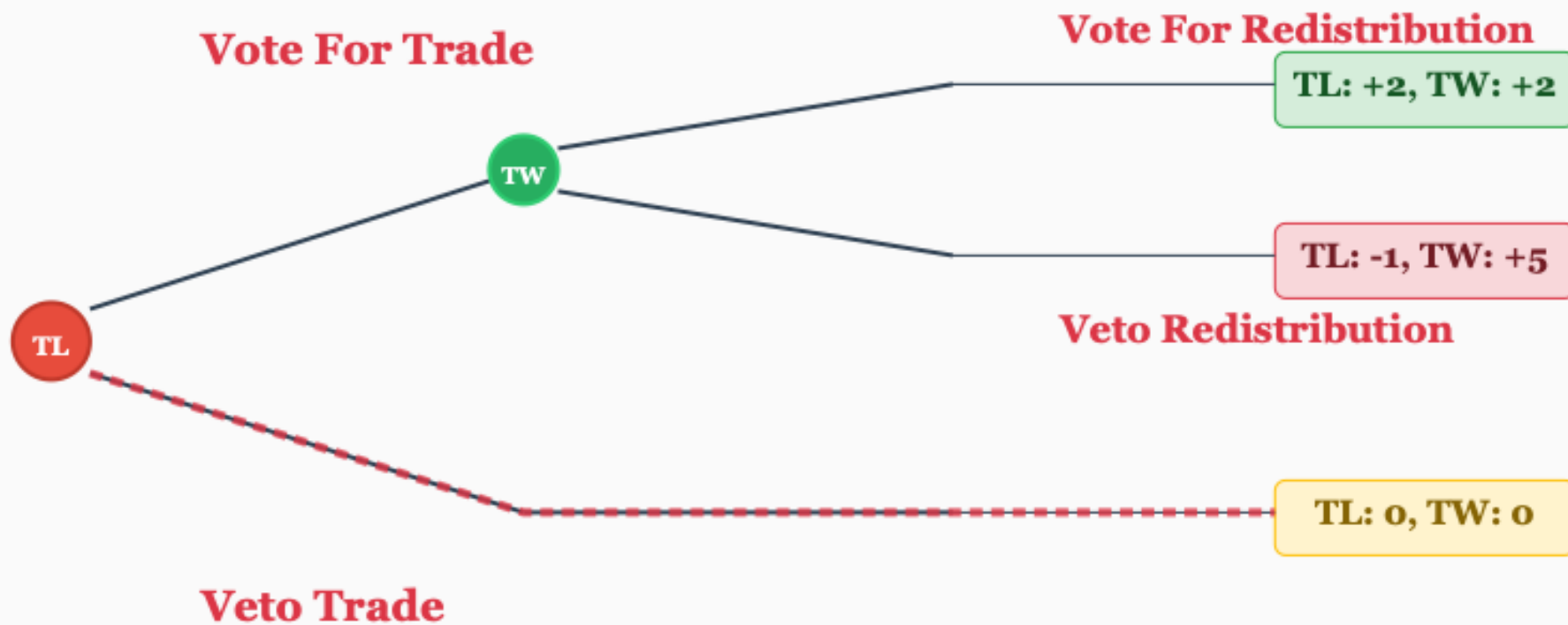
TRADE LOSER

Industry hurt by trade



TRADE WINNER

Industry helped by trade



Backward Induction: Trade Winner prefers Veto Redistribution → Trade Loser prefers Veto Trade

Solving the sequential “Gains from Trade” game

Backward induction: start at last decision node, and work your way backward to the start, assuming players maximize their payoffs & expect others to do the same.

- TL (1st mover) assumes that TW wants to maximize its own profits
 - If TL allows trade, TW will veto redistribution → TL payoff: -1
 - If TL vetoes trade, status quo preserved → TL payoffs: 0
- Anticipating that TW will block redistribution, TL is better off blocking free trade.
- Sequential Nash equilibrium: no trade (0,0).
- TW & TL would both be better off with free trade + redistribution.
- But TW cannot *credibly commit* to vote for redistribution.

Application: Property rights and investment

A main tenet of modern institutional economics: **secure property rights** are necessary for investment & economic growth.

Industrial revolution could happen in England because the bourgeoisie there enjoyed secure property rights guaranteed by the rule of law and a powerful Parliament.

- When the King had absolute power, bourgeoisie would not invest in large-scale innovative machinery for fear that any profits would then be expropriated by the King/nobility.
- After the 1688 Glorious Revolution, Parliament (controlled by bourgeoisie merchants) had primacy over the King.
- The bourgeoisie could then make industrial investments, knowing the King would not be able to expropriate the profits.





Property Rights Security and Investment

A Sequential Game Illustrating the Credible Commitment Problem



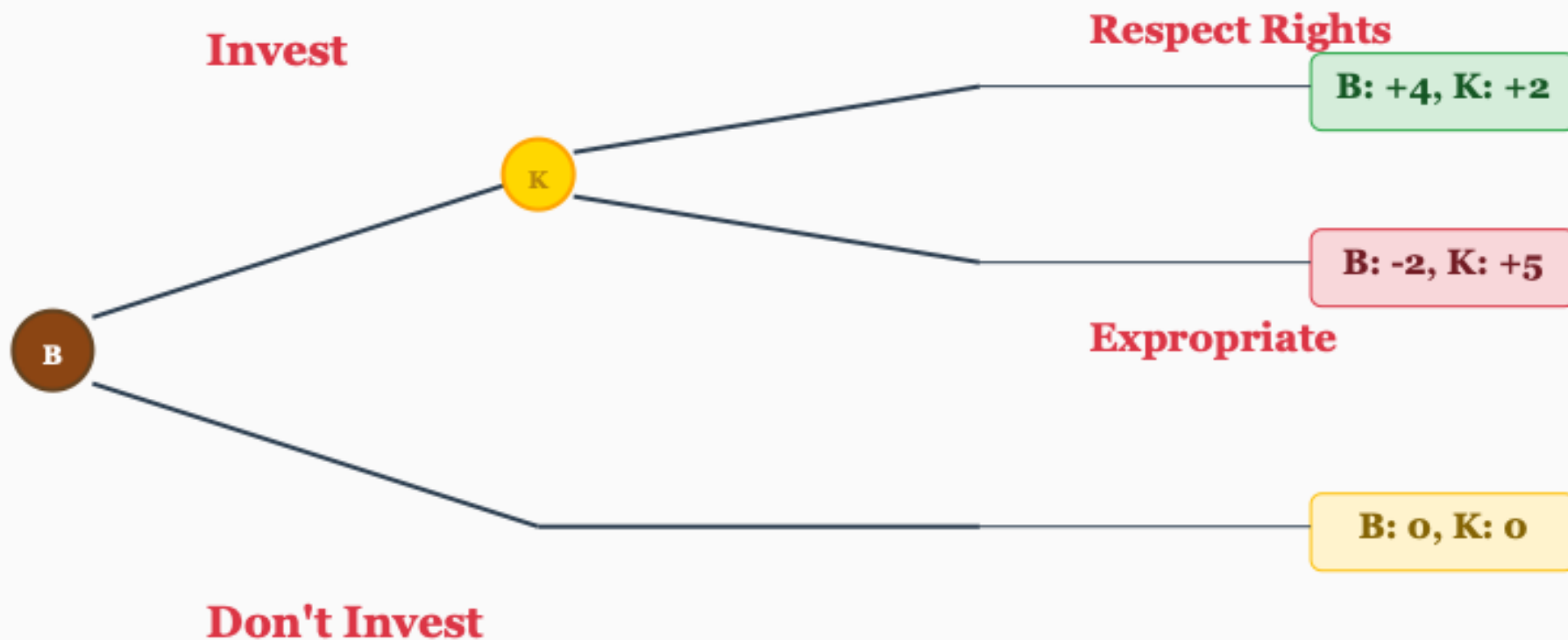
BOURGEOISIE

Merchant class with capital



KING

Sovereign ruler



Backward Induction: King prefers Expropriate → Bourgeoisie prefers Don't Invest

Application: Property rights and investment

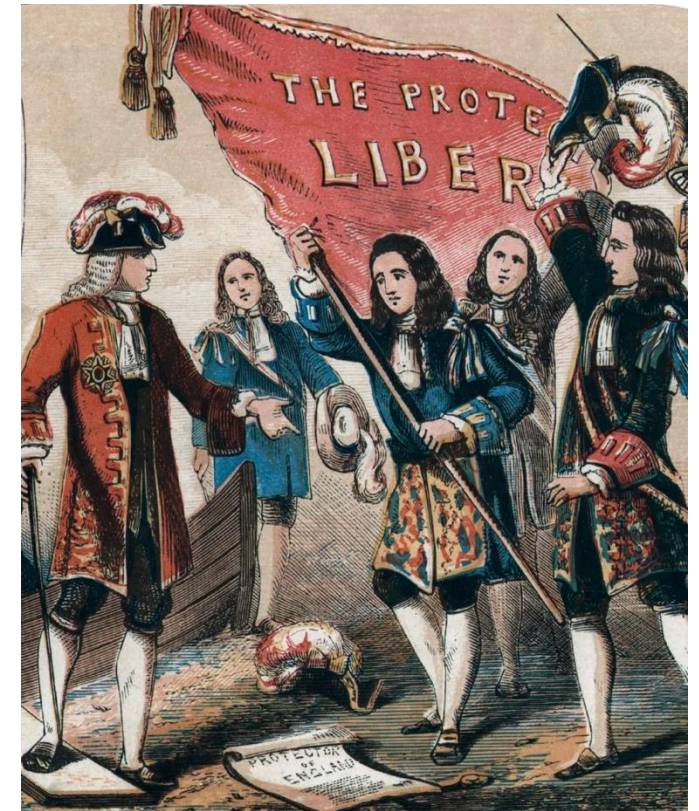
Bourgeoisie & King both better off with industrial investment and moderate taxation than in the no-investment/no-growth equilibrium.

But there is a **credible commitment problem**:

- Once the investment is done and the profits realized, nothing can stop the King from expropriating.
- An all-powerful King cannot *credibly commit* to respect the property rights of merchants & industrialists.

Only solution is for the King to give up its taxation power to a powerful Parliament controlled by the bourgeoisie.

- Glorious Revolution solved the coordination problem!



Pareto efficiency: a way of evaluating outcomes

- Coordination failures → decentralized pursuit of self-interest produces **socially inefficient outcomes**
- *But how do we determine if an outcome is socially efficient? How do we go about comparing outcomes?*
- One criterion is **Pareto efficiency**:
- **A is Pareto superior to B** if no participant prefers B over A, and at least one prefers A (*Pareto comparison*).
- **An outcome is Pareto efficient if** no other feasible outcome is Pareto superior to it.

Pareto efficiency in the fishermen dilemma

		Bob	
		Fish 10 hours	Fish 12 hours
AI	Fish 10 hours	3, 3	1, 4
	Fish 12 hours	4, 1	2, 2

1. Is the N.E. (12h, 12h) Pareto efficient?

2. Is (12h, 10h) Pareto efficient?

Pareto efficiency in the fishermen dilemma

		Bob		
		Fish 10 hours	Fish 12 hours	
AI	Fish 10 hours	3, 3	1, 4	
	Fish 12 hours	4, 1	2, 2	

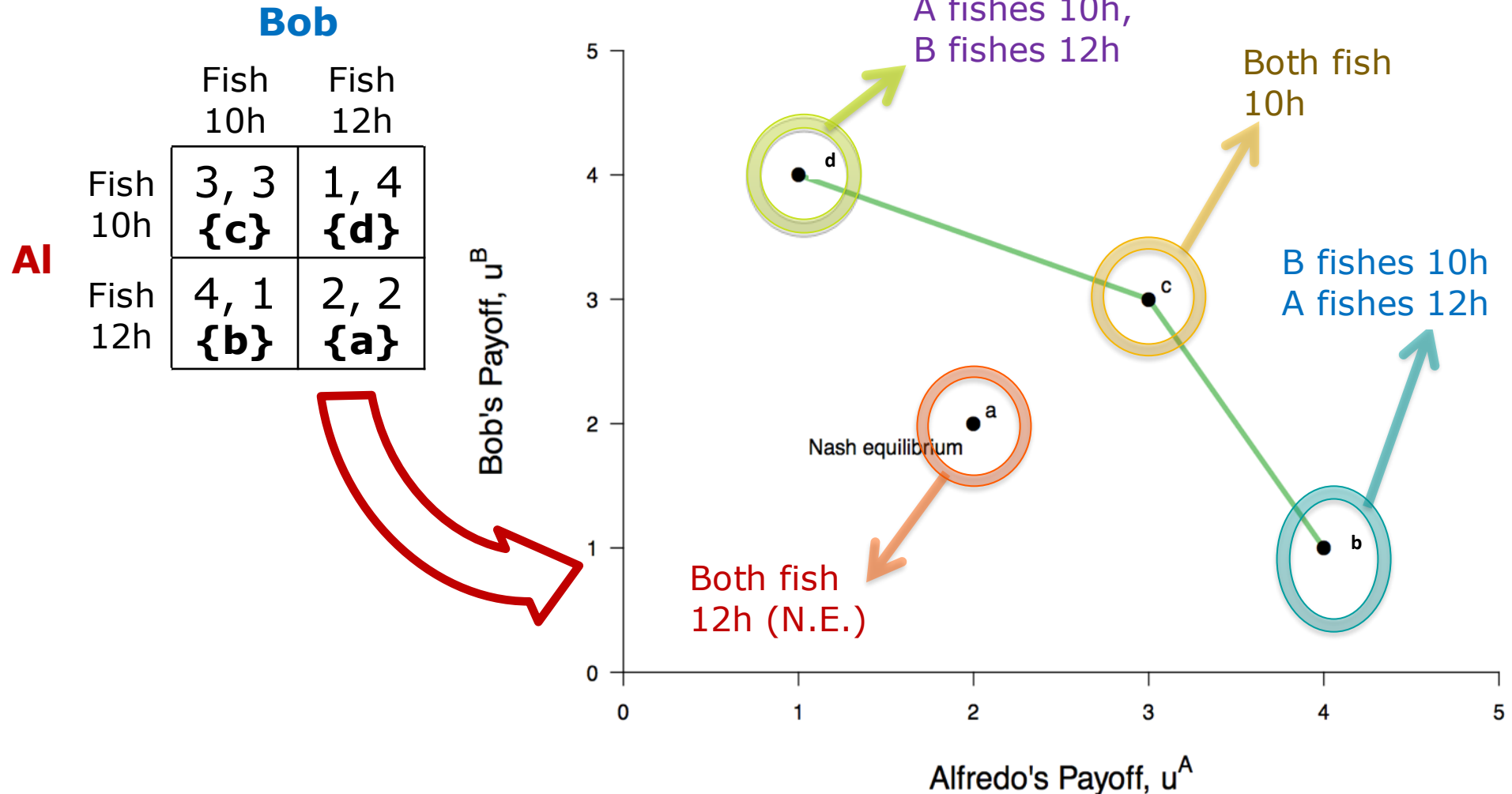
PARETO-EFFICIENT

NOT PARETO-EFFICIENT

Using a graph to visualize Pareto efficiency

- We can plot the outcomes of a game, to assess Pareto-efficiency visually.
- 1. Plot the payoff of Player A on the horizontal axis, and the payoff of Player B on the vertical axis.**
 - Each point in this graph will represent a combination of outcomes for A and B.
 - 2. Find & mark the points in the graph corresponding to each possible outcome of the game.**

This is the plot of the fishermen dilemma:



An outcome is Pareto-superior to another if it lies to the north-east of it in the plot.

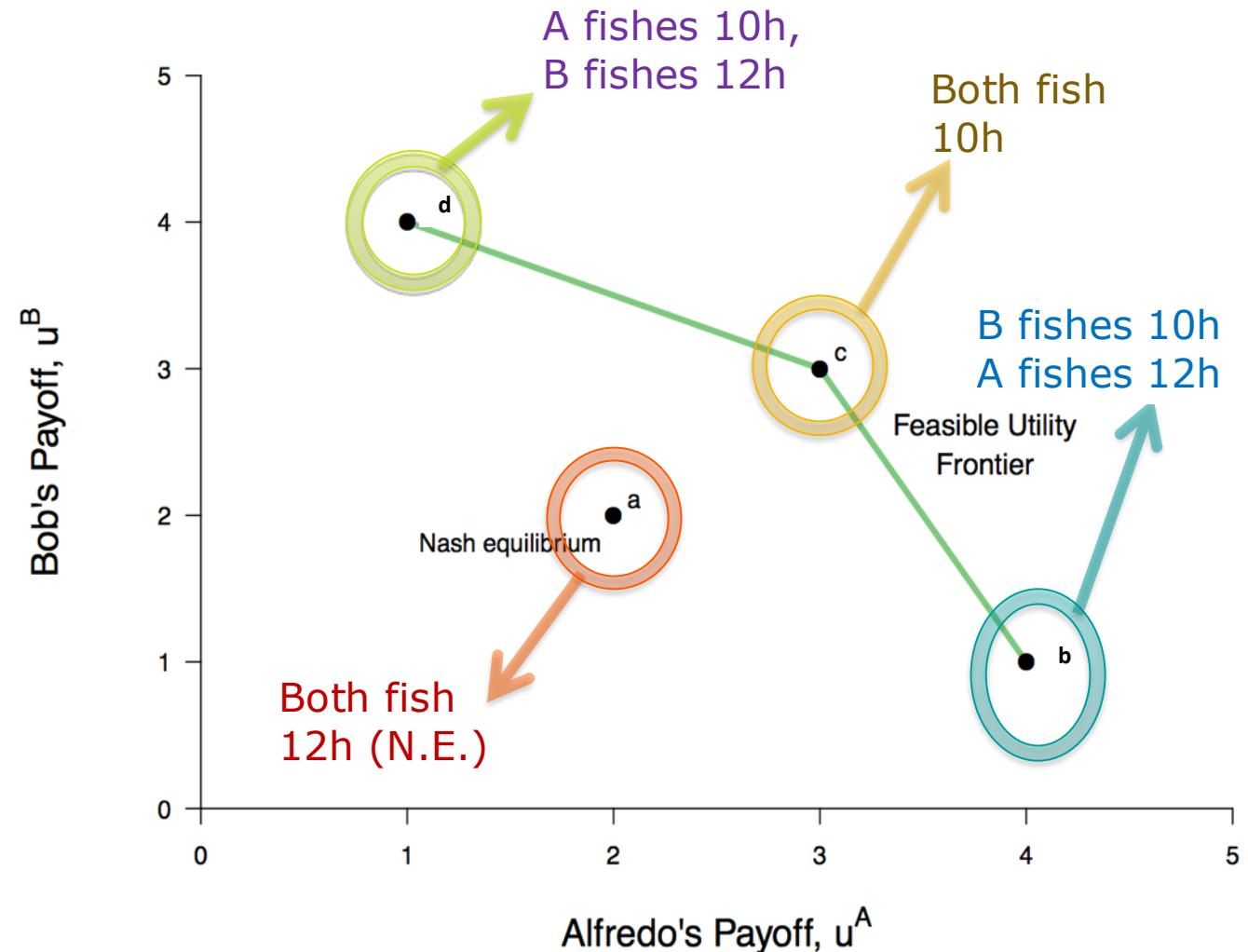
→ if it's northeast, it is better for both (or better for one and indifferent for the other)

An outcome is Pareto-efficient if there is no feasible outcome that lies to the north-east of it.

c is Pareto-superior to **a**.

d, **c** and **b** are Pareto efficient (can't find an outcome that lies north east).

a is not Pareto efficient (there is an outcome that lies north-east of it).



Strenghts of the Pareto criterion

- The Pareto criterion provides a fairly uncontroversial and objective way to identify the “bad” outcomes.
- Good at identifying outcomes that “none would have chosen”.
- If we are in a Pareto-inefficient outcome, we are foregoing an opportunity to make everyone better off (or at least someone better off and no one worse off).
- That seems bad indeed!

Shortcomings of the Pareto criterion

The Pareto criterion is at the same time too strong and too weak.

Too strong: making only Pareto-efficient changes would bring to inertia.

- *Usually, we have to go against some particular interest in order to benefit society as a whole. (ex.: taxing the super-rich to build schools). Pareto comparisons provide little info about these issues.*

Too weak: it is not able to exclude outcomes that are “bad” because inequitable or unfair.

- *Take the problem of dividing up a pie. All the Pareto criterion says is “don’t throw away any of the pie”. But it is unable to say that an outcome in which one player gets all the pie while the other starves is unfair.*

Pareto efficiency & the classical institutional challenge

Pareto efficiency is **necessary but not sufficient** to address the classical institutional challenge:

- Good institutions should generally bring about Pareto efficient outcomes (no good food should be left on the table!).
- But good rules should also address important concerns of distribution and fairness, about which the Pareto criterion can't say anything.

Coordination problems : A game-theoretical definition

A coordination problem occurs when the non-cooperative interaction of two or more agents leads to a Nash equilibrium that is not Pareto efficient, or when there are multiple N.E.

Classification of games

Based on whether they present coordination problems, and of which kind.

Invisible Hand games: unique Pareto-efficient N.E. → no coordination problem

Coordination games: decentralized N.E. not Pareto efficient, or multiple equilibria.

- Prisoners' dilemmas: no Pareto-efficient N.E. → coordination failure
- Assurance games: multiple equilibria, some inefficient → common interest
- Disagreement games: multiple equilibria, all Pareto efficient → conflict of interest

Consider the following game

Arkady & Barbara decide whether to plant corn or tomato

Arkady is better at growing tomato

Barbara is better at growing corn

		Barbara	
		Corn	Tomato
Arkady	Corn	1, <u>2.5</u>	2, 2
	Tomato	<u>4</u> , <u>4</u>	<u>2.5</u> , 1

1) What is the Nash Equilibrium?

2) Is the N.E. Pareto efficient?

Invisible Hand Game

Invisible Hand Game:

There is a unique N.E. & it is Pareto efficient.

Uncoordinated individual actions lead to socially efficient outcomes.

		B	
		Corn	Tomato
A	Corn	1, <u>2.5</u>	2, 2
	Tomato	<u>4</u> , <u>4</u>	<u>2.5</u> , 1

The “planting in Palanpur” game

1) What is/are the N.E.?

3) Are the two N.E. Pareto efficient?

		Bina	
		Plant Early	Plant Late
Aram	Plant Early	<u>4</u> , <u>4</u>	0, 3
	Plant Late	3, 0	<u>2</u> , <u>2</u>

Assurance game

Coordination game: there is a Pareto inefficient N.E.

Specifically, **Assurance game:** *players agree on which outcome would be best, but they need coordination to get there.*

		B	
		Plant Early	Plant Late
A	Plant Early	4, 4	0, 3
	Plant Late	3, 0	2, 2

Two defining characteristics of Assurance Games:

- Multiple equilibria that are Pareto rankable
one is Pareto-superior to the other(s).
- Strategic complementarity (or positive feedback effects):
the payoff from a strategy is increasing in the number of players adopting it—you want to do what others are doing.

The language game

Ben
(English-speaker)

Swahili English

Aisha
(Swahili-speaker)

Swahili	4, 2	0, 0
English	0, 0	2, 4

1) What is/are the N.E.?

2) Are the two N.E.
Pareto efficient?

Disagreement Game

The “Language game” is a coordination game (multiple equilibria)

Specifically, a disagreement game: multiple equilibria which are all Pareto efficient.

		Ben (English-speaker)	
		Swahili	English
Aisha (Swahili-speaker)	Swahili	4, 2	0, 0
	English	1, 1	2, 4

In disagreement games:

- you cannot move from one N.E. to another without hurting somebody
- Players disagree on which NE is best.

Players agree they should *coordinate*:
you want to speak the same language as the other (strategic complementarity)

But they are in *conflict* over which outcome to coordinate on (Swahili or English?) → **conflict of interests**.

Strategic complementarities

- Sometimes, payoff from an action is higher if others are choosing that action as well.
 - *Any examples in your life?*
 - Communication platforms/social media: You want to be on Whatsapp only if your friends use it too.
- Strategic complementarities often give rise to multiple Nash equilibria
- *Path dependence*: Historical patterns tend to persist.
- The same individuals, playing the same game, can end up in different equilibria depending on the initial situation.

Institutions & Coordination Problems

- Coordination problems = Pareto-inefficient Nash Equilibria or multiple equilibria.
- To solve them, we need to change the *payoff structure* of the game
 - ‘payoff structure’ just means “what payoff each player gets in each possible outcome”.
- We must change the rules of the game.
- The rules of the game are determined by *institutions*: laws, social norms, mutual expectations...



Institutions & Coordination Problems

Institutions are the rules of the game that determine the outcome of coordination problems.

Example: Why do you drive on the left in the UK?

Because otherwise you will be arrested (*laws*).

Because otherwise other people will think you are nuts or despise you (*social norms*).

Because you expect everyone else to drive left, so it is in your best interest to do the same (*mutual expectations*).



Institutions & Coordination Problems

By changing the rules of the game, institutions can address coordination problems.

Two possible problems to be solved:

- How to get there? If there is a Pareto efficient N.E., but we are stuck in an inefficient one (e.g. Assurance Game). Then we need a coordination device for people to converge on the efficient equilibrium.
- How to stay there? If the efficient outcome is not a N.E. (e.g. Prisoner Dilemma), we need to change the payoff structure to make it a N.E.

How to 'break' a Prisoners Dilemma

- Prisoners' dilemma: a unique and inefficient N.E.
- How can the right institutions transform the payoff matrix to solve the coordination problem?
- Take the fishermen dilemma.
- *Liability rule*: if you defect (fish 12h), you must compensate the other fisher for the harm that you are causing to them.
- This *legal* institution can turn a Prisoners Dilemma game (like the fishermen dilemma) into an Invisible Hand game.

The fishermen dilemma with a liability rule

AI

		Bob	
		Fish 10h	Fish 12h
Fish 10h		3, 3	1+2, 4-2
Fish 12h		4-2, 1+2	2, 2

What is the N.E.
with the liability
rule?

The fishermen dilemma with a liability rule

AI

		Bob	
		Fish 10h	Fish 12h
Fish	10h	3, 3 <u>3</u>, <u>3</u>	1+2, 4-2 <u>1</u>, 4-2
	12h	4-2, 1+2 4-2, <u>1</u>+2	2, 2

- This re-definition of **property rights** succeeded!
- We have **internalized** the externalities, and now the unique Nash Equilibrium is Pareto efficient.
- A simple example of how **institutions can address coordination problems**.
- Of course things aren't always that easy: here we assumed that behavior is observable and that contracts are costless to enforce!



Thank you for your attention

The Public Goods Game (PGG)

- Goods can be of 4 types:

	<i>Excludable</i>	<i>Non-excludable</i>
Rival	Private good (clothing, food)	Common property (Pool) Resource (fishing stocks, potential buyers)
Non-rival	Club good (streaming music, online movies)	Public good (global climate, rules of calculus)

- ‘Public goods’ are non-excludable and non-rival.
 - *Non-excludable*: hard to stop people from accessing it.
 - *Non-rival*: one person's use doesn't reduce availability for others.
- Classic example: a lighthouse.
- Also: clean air, knowledge, global climate, etc.

Public Goods & Coordination Failures

- Non-excludability means it is very hard to make profits by producing a public good.
- As a result, no one has an incentive to provide them.
- *Underprovision* without institutional intervention.
- *Coordination failure* due to (positive) *externality*.

The Public Goods Game (PGG)

- n players
- each starts with endowment z
- player i 's contribution to common pot: e^i
- player i keeps for herself : $z - e^i$
- public good returns = sum of contributions $(\sum_i e^j)$ * rate of return (M)
- $0 < M < 1$.
- *Own payoff = Endowment – Contribution + Public good returns*
- $u^i = z - e^i + M (\sum_i e^j)$

Your turn:

- Write the payoff matrix of the public good game with:
 - 2 players ($n=2$)
 - Initial endowment $z=10$
 - Rate of return $M=0.8$
 - 2 possible choices: $e^i = 10$ or $e^i = 0$.
- Then find the Nash equilibrium

		Player 2	
		$e^2 = 10$	$e^2 = 0$
Player 1	$e^1 = 10$	? , ?	? , ?
	$e^1 = 0$	? , ?	? , ?

Your turn:

- Write the payoff matrix of the public good game with:
 - 2 players ($n=2$)
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 - 2 possible choices: $e^i = 10$ or $e^i = 0$.
- Then find the Nash equilibrium

		Player 2	
		$e^2 = 10$	$e^2 = 0$
Player 1	$e^1 = 10$	16, 16	8, 18
	$e^1 = 0$	18, 8	10, 10

Player 1

$e^1 = 10$

$e^1 = 0$

Player 2

$e^2 = 10$

$e^2 = 0$

16, 16	8, <u>18</u>
<u>18</u> , 8	<u>10</u> , <u>10</u>

Under self-interest,
the (inefficient) Nash
Equilibrium is both
contributing 0.

A Prisoners' Dilemma.

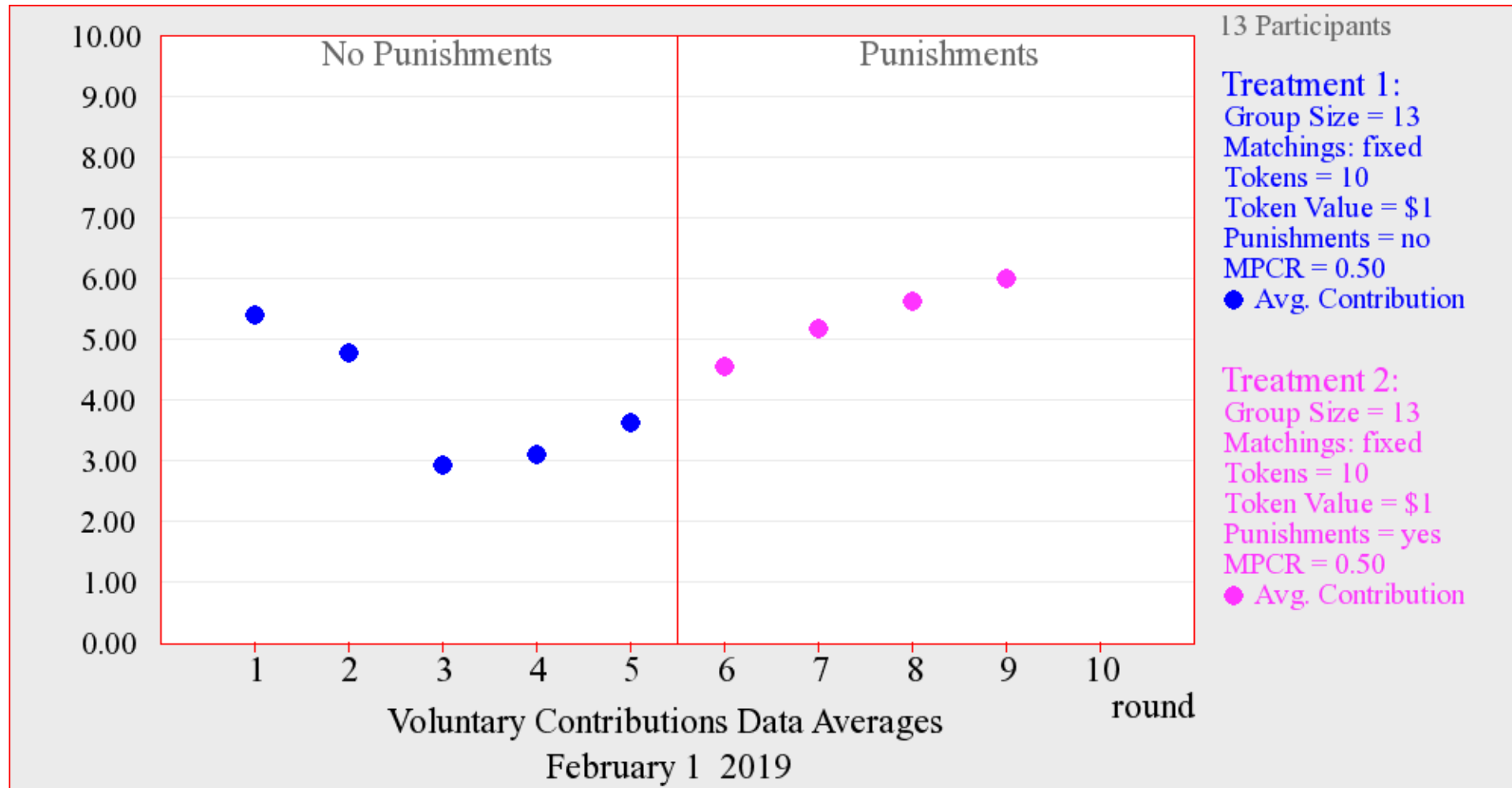
Public Goods Game (PGG)

- Self-interest Nash Equilibrium predicts zero contributions;
 - For each \$ you contribute, you get $M < 1$ back;
- Lab experiments with the PGG:
 - One-shot-game: contributions around $\frac{1}{2}$
 - Repeated game: contributions start high, then decline towards zero;
- How to explain the decline in contributions?
 - Self-interest + learning?
 - Reciprocity?

The Public Goods Game with punishment

- A Public Goods Game, BUT after each round
 - contributions are made public;
 - each player can punish some other (reduce their payoff).
 - but need to pay a cost to 'punish'.
- Empirical results from PGG w/ punishment:
 - non-cooperators get punished.
 - contributions remain high throughout all rounds.
 - proof of *reciprocity* .
 - reciprocators can keep cooperating with each other while punishing the non-cooperators.

Students from a previous class playing the public good game in a seminar:



You playing the game in the seminar today in groups of 4
(The graph shows average contributions across all groups in each round)

